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25 June 1979
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MILITARY SPECIFICATION

ROPE, EXTRA-HIGH-STRENGTH WIRE, 6 BY 19 WITH 7 BY 7 INDEPENDENT WIRE ROPE CENTER (FOR AIRCRAFT LAUNCHING AND RELATED APPLICATIONS)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of extra-high-strength wire rope, 6 by 19 having a 7 by 7 independent wire rope core for aircraft launching and related applications (see 6.1).

1.2 Classification.

1.2.1 Grade. The wire rope shall be furnished in one grade only.

1.2.2 Sizes. The wire rope shall be furnished in the sizes shown in Table II, as specified.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center (Code 93), Engineering Specifications and Standards Department, Lakehurst, NJ 08733, by using the Self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4010

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SPECIFICATIONS

MILITARY

MIL-B-121	-Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-16173	-Corrosion Preventive Compound, Solvent Cutback, Cold Application

STANDARDS

MILITARY

MIL-STD-129	-Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8-78	-Methods of Tension Testing of Metallic Materials

(Copies of ASTM Publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. The steel used in the manufacture of the wire rope covered by this specification shall be of high quality and suitable for the purpose. The wire shall be cylindrical and smooth and of uniform high quality. The wires shall be uncoated i.e., not galvanized, tinned, etc. The rope shall not contain any non-metallic filler material.

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3.2 Design and construction. The design and construction of the wire rope shall be in accordance with this specification. No deviation from any of the requirements of this specification shall be permitted unless specifically authorized in writing by the procuring activity through the contracting officer. Copies of any letter requesting a deviation shall be forwarded to the Commander, Naval Air Systems Command (Code AIR-551), Washington, DC 20361 and to the Commanding Officer, Naval Air Engineering Center (Code 91), Lakehurst, New Jersey 08733.

3.2.1 Rope construction. The rope shall be 6 by 19 construction, right hand helix regular lay, preformed with a flexible metallic core. The rope shall have one core, and six strands of 19 main wires and six filler wires each, making a total of 114 main wires and 36 filler wires in the outer strands. The core shall be of 7 by 7 construction and fabricated of the same materials as the outer strands.

3.2.2 Strand construction. Each of the six strands of the wire rope shall have a main wire center, six main wires in the inner layer, and 12 main wires in the outer layer. There shall be six filler wires between the inner layer and the outer layer. All the main wires in the inner layer shall be one diameter (3.2.2.1). All the main wires in the outer layer shall be one diameter. All the filler wires shall be one diameter. The strand shall be fabricated in one operation.

3.2.2.1 One diameter. Wires in a wire rope shall be considered to be of one diameter if the difference between the diameter of the smallest wire and the diameter of the largest wire is not greater than the value specified in Table I.

TABLE I

WIRES OF ONE DIAMETER

Wire Diameter	Difference Between the Diameter of the Smallest Wire and the Diameter of the Largest Wire
Inch	Inch
Up to 0.029	0.0015
0.030 to 0.059	0.0020
0.060 to 0.099	0.0025
0.100 to 0.149	0.0030

3.2.2.2 Pitch. All the wires in a strand fabricated in one operation shall have the same pitch, and each wire shall be in contact with the same adjacent wires for its entire length.

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3.2.3 Joints. The strands of the wire rope shall contain no joints made after fabrication. If joints are made in the individual wires before the fabrication of the strands, the distance between a joint in any one wire and the joint in any other wire in the same strand shall be not less than 18 inches.

3.2.4 Rope properties. The wire rope shall have the properties specified in Table II.

3.2.5 Seizing. Each end of each length of wire rope shall be suitably seized before each cutting operation is performed.

3.2.6 Preforming. The individual wires and strands shall be preformed into the helical shape they will have in the finished wire rope.

3.3 Ductility. Wires taken from the completed wire rope shall not fracture when tested by the method specified in 4.3.2.2.

3.4 Length. The length in which the wire rope is to be furnished shall be as specified in the contract or order. Not more than one size or length of wire rope shall be furnished on a single reel which shall be identified as a shipping reel.

3.5 Lubrication. A suitable non-corrosive lubricant shall be applied to the strands during fabrication of the wire rope. When special lubricants, for use at high temperature or other unusual applications, are required, they shall be as specified in the contract or purchase order.

TABLE II

WIRE ROPE PROPERTIES

Diameter			Max Permissible Difference in Diameter when seizing is removed	Approx. Weight Per Foot	Breaking Strength 1/		Max Pitch
Nominal	Min	Max			Nominal	Min	
Inches	Inches	Inches	Inch	Pounds	Pounds	Pounds	Inches
11/16	43/64	45/64	1/32	0.78	49500	48200	4.5
7/8	7/8	59/64	3/64	1.35	78500	76500	5.7
1	1	1-3/64	3/64	1.76	102000	99500	6.5
1-1/8	1-1/8	1-11/64	3/64	2.23	128000	125000	7.3
1-1/4	1-1/4	1-5/16	1/16	2.75	157000	153000	8.1
1-3/8	1-3/8	1-7/16	1/16	3.33	188000	184000	8.9
1-1/2	1-1/2	1-9/16	1/16	3.96	224000	218000	9.8
1-5/8	1-5/8	1-23/32	3/32	4.65	260000	254000	10.6
1-3/4	1-3/4	1-27/32	3/32	5.39	301000	294000	11.4
1-7/8	1-7/8	1-31/32	3/32	6.19	338000	329000	12.2
2	2	2-3/32	3/32	7.04	376000	367000	13.0

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1/ Nominal breaking strength values are to be used for design purposes. Minimum breaking strength values are acceptable when testing rope under this specification to account for slight reduction in strength resulting from the use of short test lengths.

3.6 Workmanship. The wire rope shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability. The wire shall be free from splits, cold shuts, or other detrimental defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. *Quality conformance* inspection of the wire rope shall consist of all the examinations and tests under this specification.

4.2.1 Lot formation. All individual shipping reels of wire rope coming from the same master reel shall constitute a lot. A master reel shall consist of wire rope made only from continuous lengths of strands laid helically around a core.

4.2.2 Quality conformance inspection sampling. One 2-1/2 foot sample and one 7 foot sample shall be taken from the master reel to represent each reeled length of wire rope. Where a shipping reel cannot be definitely identified with a master reel which has been tested and accepted, the individual shipping reel shall have samples selected as above and tested. The preforming test and the ductility test shall be made on the short sample. The long sample shall be pulled to destruction to determine its breaking strength.

4.3 Methods of examination and test.

4.3.1 Examination of product. The wire rope shall be examined throughout its length to determine conformance with the requirements of this specification not covered by tests. For this purpose, the rope shall travel no faster than 100 feet per minute and shall be stopped for closer examination when required. This inspection shall take place at the time the rope is wound on shipping reels.

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4.3.2 Test methods.

4.3.2.1 Preforming test. The diameter at the end of a wire rope specimen shall be measured. The seizing adjacent to the cut shall be removed and the diameter shall be measured again. The increase in diameter due to the removal of the seizing shall not be more than the values given in Table II.

4.3.2.2 Ductility. From each short sample (4.2.2) not less than one specimen of each size of main wire from each strand shall be selected and the ductility determined. For wire rope one inch in diameter or smaller, the ductility shall be determined by the kink test (4.3.2.2.1) while for wire rope larger than one inch diameter, the ductility shall be determined by the wrapping test (4.3.2.2.2).

4.3.2.2.1 Kink test. The ductility shall be determined by the ability of the wire specimens to withstand without fracture, one tight kink. The tight kink shall be made by bending the wire into a loop and then pulling the loop tight until the opening is no larger than the diameter of the wire. The loop shall then be opened so that the wire ends are straight and at right angles to each other.

4.3.2.2.2 Wrapping test. The ductility shall be determined by the ability of the wire specimens to withstand without fracture, wrapping in a close helix for six complete turns around a mandrel. The diameter of the mandrel shall be the same as the diameter of the wire.

4.3.2.3 Breaking strength. The breaking strength of the wire rope shall be determined on the long sample (4.2.2). The wire rope shall be tested in accordance with the requirements of ASTM E8-78. The breaking strength shall meet the requirements specified in Table II. The rope ends shall be socketed in zinc or other suitable metal.

4.4 Contractor's inspection reports. The contractor shall furnish test reports showing quantitative results for all tests required by this specification in accordance with the requirements specified in the contract or purchase order.

5. PACKAGING

5.1 Preservation. The wire rope shall be protected from corrosion during shipment and storage by preserving with a compound conforming to Specification MIL-C-16173, Grade 1. The compound shall be applied over the entire length of the rope prior to final reeling. It shall cover all exposed wire surfaces and penetrate the crevices between wire and strands. The compound shall be of a sufficiently tacky consistency that it will not drain off after extended storage.

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5.2 Packing. Wire rope shall be wound on substantial commercial type reels. The outer periphery of the reels shall be covered with grease-proof barrier material conforming to Specification MIL-B-121, Type II, Grade A, Class 2. The outside diameter of each reel shall be enclosed with solid blocked wood sheathing. Only one length of rope shall be furnished on any single reel. The reels shall be capable of withstanding storage, rehandling, and reshipment without necessity of repacking.

5.2 Marking. Reels shall be marked in accordance with Standard MIL-STD-129.

5.3.1 Additional marking. The following additional marking shall also be added:

a. Wire rope, steel, 6 by 19 with 7 by 7 independent wire rope center, Specification MIL-R-7871

b. Shipping reel number.

c. Length.

d. All shipping reels shall be marked with the words FOR AIRCRAFT LAUNCHING AND RELATED APPLICATIONS.

e. Each shipping reel shall have attached to it a test certificate indicating the master reel from which it was taken and the test results obtained on the master reel.

6. NOTES

6.1 Intended use. Wire rope covered by this specification is intended primarily for use in aircraft launching but may also be used in other critical applications. This wire rope was designed with a minimum margin of safety and is subjected to impact and dynamic loads. Failure of this wire rope could result in the loss of lives of both aircraft and flight deck personnel as well as loss of an aircraft. The importance of providing a product of uniformly excellent quality cannot by overemphasized.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Nominal size (diameter)

c. Quantity of wire required

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6.3 Definitions.

6.3.1 Wire. Each individual cylindrical steel rod or thread is designated as a wire.

6.3.2 Strand. Each group of wires helically twisted or laid together is designated as a strand.

6.3.3 Wire rope. A group of strands helically twisted or laid about a central core is designated as a wire rope. The strands and the core act as a unit.

6.3.4 Preformed type. Wire rope in which the strands and their wires are permanently shaped during fabrication to the spiral form they assume in the finished wire rope shall be identified as preformed wire rope.

6.3.5 Diameter. The diameter of the wire rope is the diameter of the circumscribing circle.

6.3.6 Lay or helix. The helical form taken by the wires in the strand and by the strands in the wire rope is characterized as the lay or helix of the strand or wire rope, respectively.

6.3.7 Right hand helix. In a right hand helix wire rope the strands form a helix about the axis of the rope similar to the threads of a right hand screw.

6.3.8 Regular lay. In a regular lay wire rope the direction of the helix of the strands is opposite the direction of the helix of the rope.

6.3.9 Pitch (or length of lay). The distance, parallel to the axis of the strand or wire rope, in which a wire or strand makes one complete turn about that axis is designated as the pitch (or length of lay) of the strand or wire rope, respectively.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the changes.

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